

```
void fun1(struct node* head)
{
    if(head == NULL)
        return;
    fun1(head->next);
    printf("%d ", head->data);
}
```

- A. Prints all nodes of linked lists
- B. Prints all nodes of linked list in reverse order
- C. Prints alternate nodes of Linked List
- D. Prints alternate nodes in reverse order

View Answer

Ans : B

Explanation: fun1() prints the given Linked List in reverse manner. For Linked List 1->2->3->4->5, fun1() prints 5->4->3->2->1.

2. A linear collection of data elements where the linear node is given by means of pointer is called?

- A. linked list
- B. node list
- C. primitive list
- D. None of these

View Answer

Ans : A

Explanation: A linear collection of data elements where the linear node is given by means of pointer is called linked list.

3. What is the time complexity to count the number of elements in the linked list?

- A. $O(1)$
- B. $O(n)$
- C. $O(\log n)$
- D. None of the mentioned

View Answer

Ans : B

Explanation: To count the number of elements, you have to traverse through the entire list, hence complexity is $O(n)$.

4. What would be the asymptotic time complexity to add a node at the end of singly linked list, if the pointer is initially pointing to the head of the list?

- A. $O(1)$
- B. $O(n)$
- C. $\theta(n)$
- D. $\theta(1)$

View Answer

Ans : C

Explanation: No Explanation.

5. What is the output of following function for start pointing to first node of following linked list? 1->2->3->4->5->6

```
void fun(struct node* start)
{
    if(start == NULL)
        return;
    printf("%d ", start->data);
    if(start->next != NULL )
        fun(start->next->next);
    printf("%d ", start->data);
}
```

- A. 1 4 6 6 4 1
- B. 1 3 5 1 3 5
- C. 1 2 3 5
- D. 1 3 5 5 3 1

View Answer

Ans : D

Explanation:fun() prints alternate nodes of the given Linked List, first from head to end, and then from end to head. If Linked List has even number of nodes, then skips the last node.

6. What is the functionality of the following piece of code?

```
public int function(int data)
{
    Node temp = head;
    int var = 0;
    while(temp != null)
    {
        if(temp.getData() == data)
        {
            return var;
        }
        var = var+1;
        temp = temp.getNext();
    }
    return Integer.MIN_VALUE;
}
```

- A. Find and delete a given element in the list
- B. Find and return the given element in the

- B. Find and return the given element in the list
- C. Find and return the position of the given element in the list
- D. Find and insert a new element in the list

View Answer

Ans : C

Explanation: When temp is equal to data, the position of data is returned.

7. Linked lists are not suitable to for the implementation of?

- A. Insertion sort
- B. Radix sort
- C. Polynomial manipulation
- D. Binary search

View Answer

Ans : D

Explanation: Linked lists are not suitable to for the implementation of Binary search.

8. In the worst case, the number of comparisons needed to search a singly linked list of length n for a given element is

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- A. $\log_2 n$
- B. $n/2$
- C. $\log_2 n - 1$
- D. n

View Answer

Ans : D

Explanation: In the worst case, the element to be searched has to be compared with all elements of linked list.

9. Which of these is an application of linked lists?

- A. To implement file systems
- B. For separate chaining in hash-tables
- C. To implement non-binary trees
- D. All of the mentioned

View Answer

Ans : D

Explanation: Linked lists can be used to implement all of the above mentioned applications.

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View Answer

Ans : D

Explanation: Linked lists can be used to implement all of the above mentioned applications.

10. In circular linked list, insertion of node requires modification of?

- A. One pointer
- B. Two pointer
- C. Three pointer
- D. None

View Answer

Ans : B

Explanation: In circular linked list, insertion of node requires modification of Two pointer.